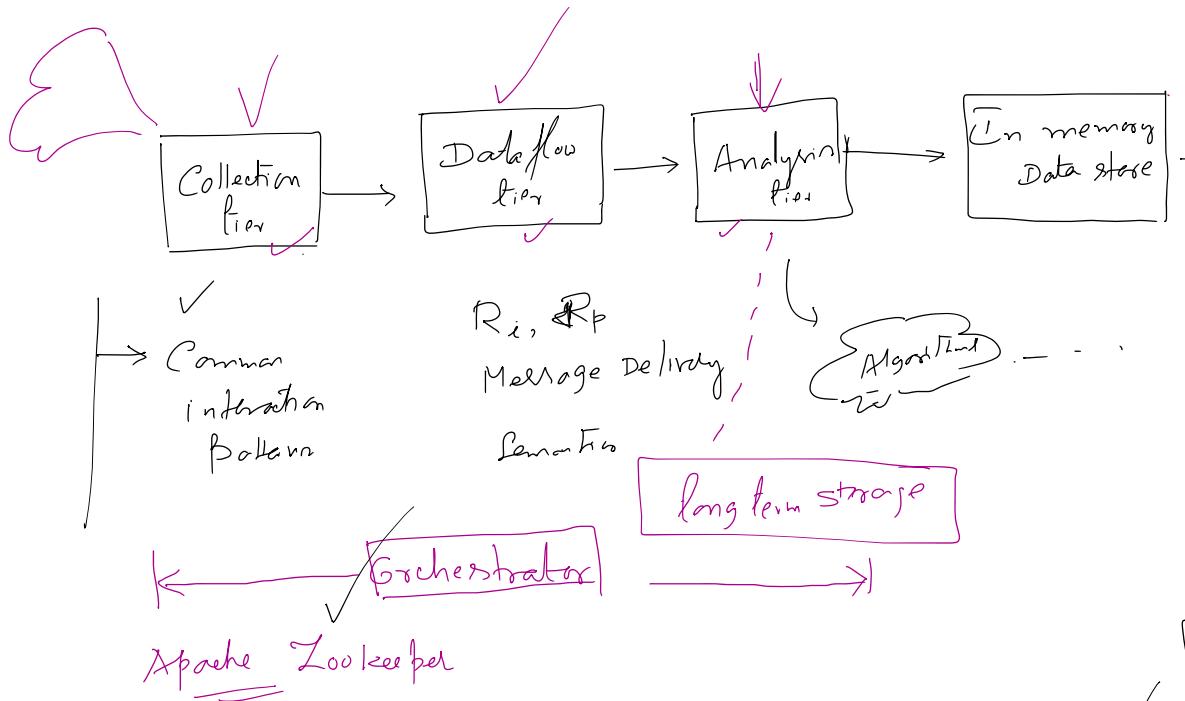
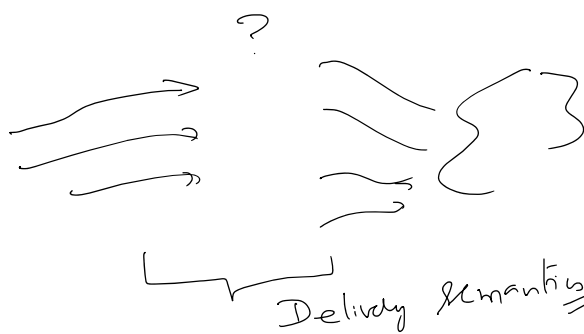


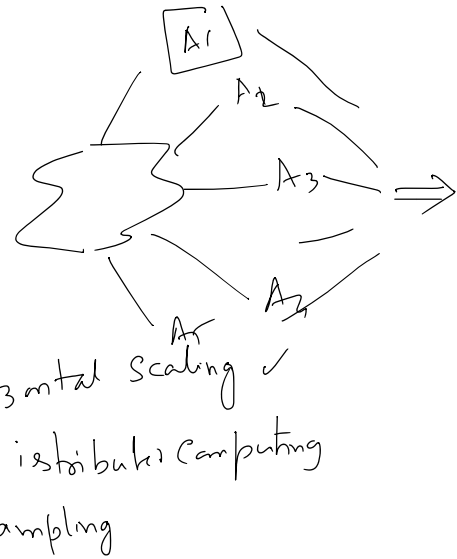


$R_i \leq R_p$: In memory (Buffer)

$R_p \ll R_i$: $R_i = 3R_p$ $R_i = 5R_p$



Message Delivery Semantics



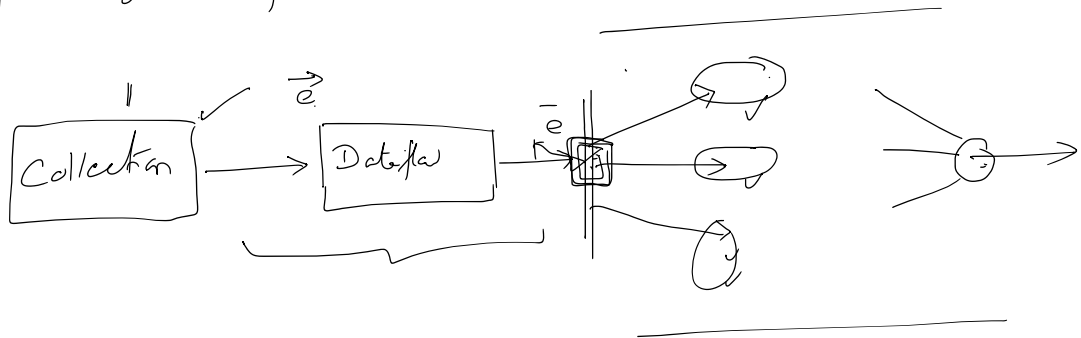
① Horizontal Scaling ✓
② Distributed Computing
Sampling

① At most once Delivery Semantics:

Message may get lost; But it will never be re-read by consumer

② At least once: Message will never be lost; But it might be re-read by consumer

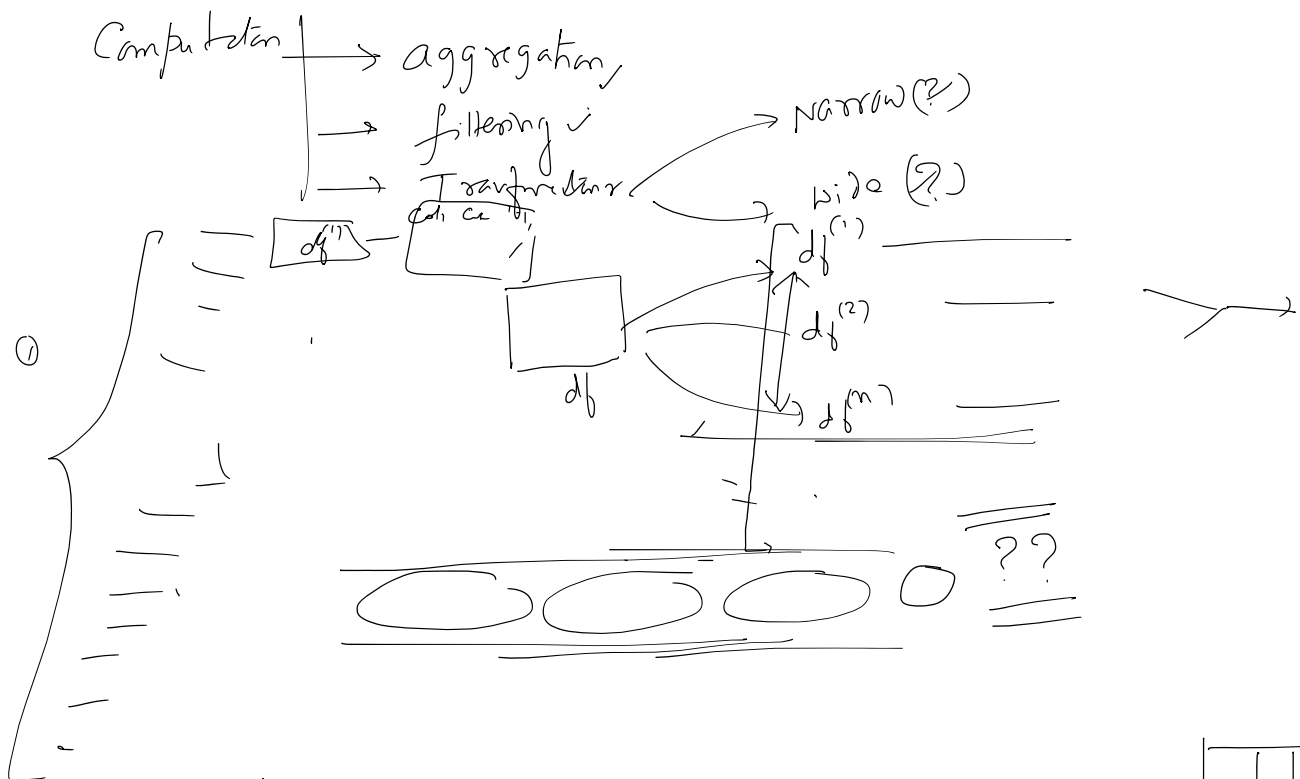
- ② At least once: Message will never be lost; But it might be re-read by consumer
- ③ Exactly once: A message is never lost; it is read by consumer once and only once



How does it ensure it works?

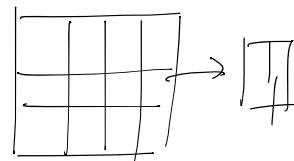
$$\psi(\vec{x}) = \varphi_1(\vec{x}') \varphi_2(\vec{x}') + \varphi_3(\vec{x}') \varphi_4(\vec{x}') + \dots$$

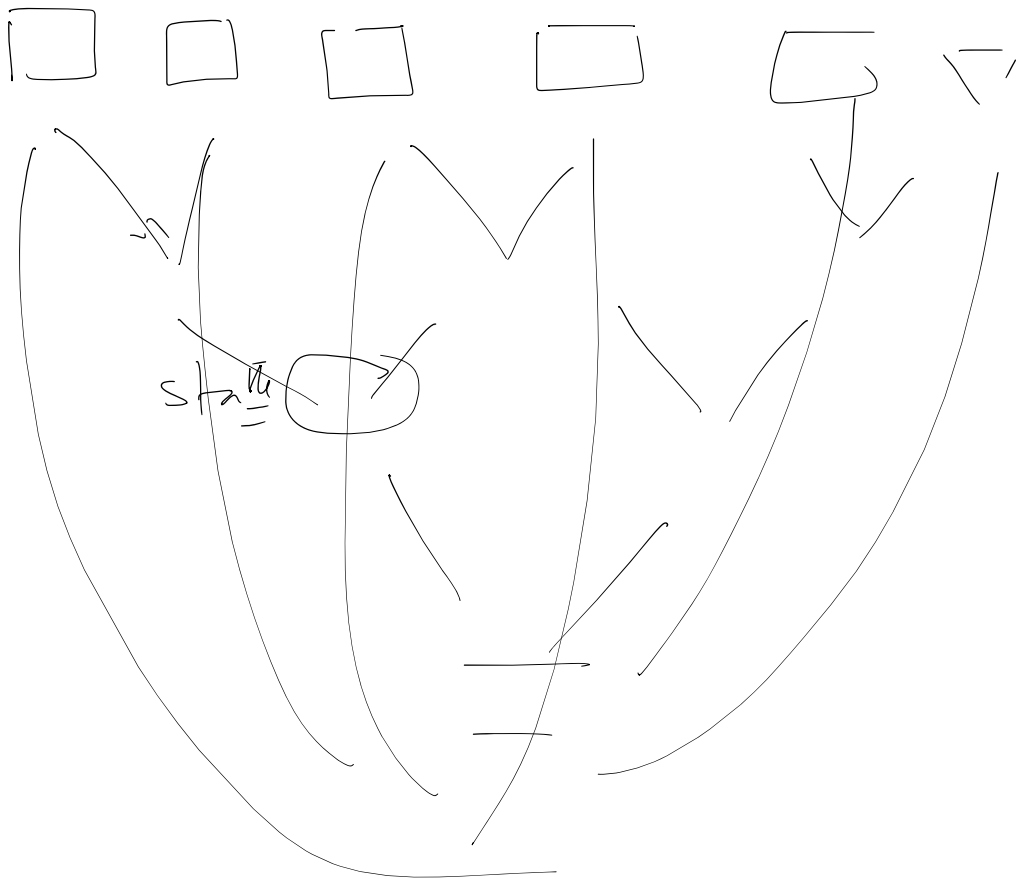
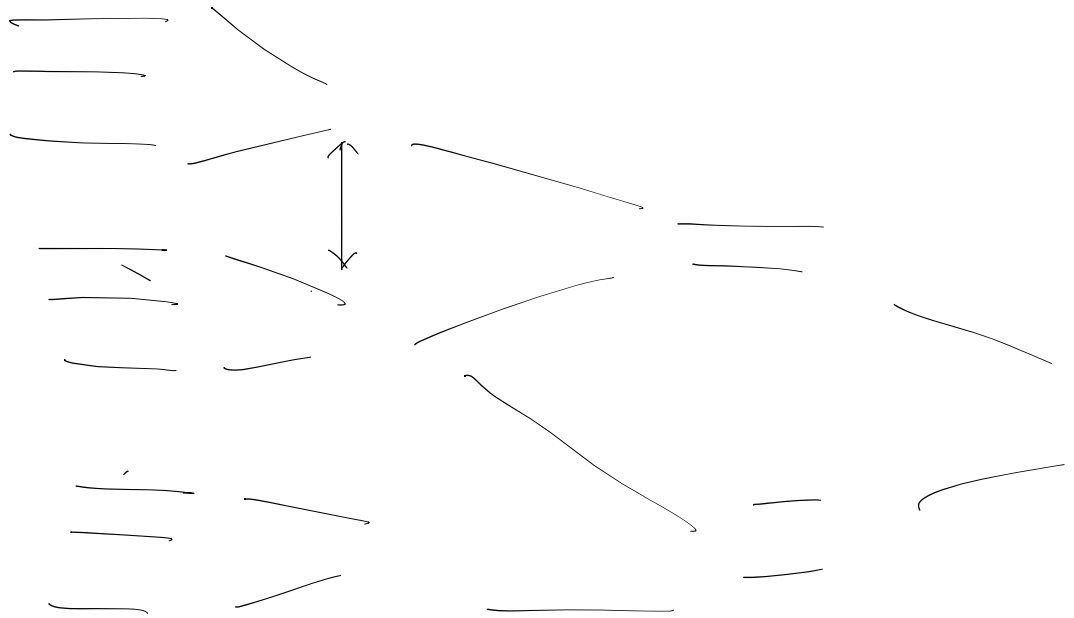
If a computation can be expressed (modelled) as
 Sum of products of individual (node level) computation,
 then it can be programmed as part of Distributed Computing



Aspects of Consideration

groupBy([col₁, col₂]).agg(col₃)





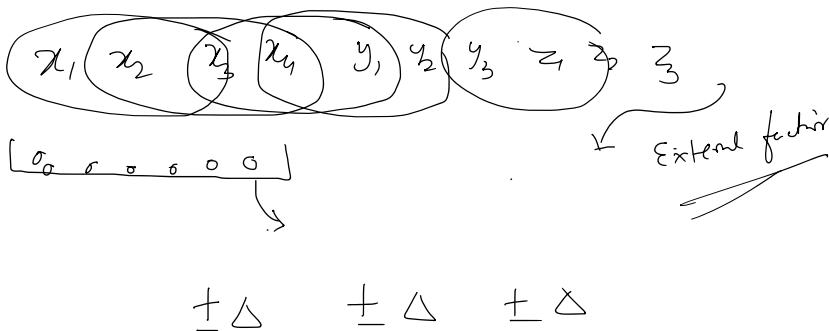
id Key

5	6	7	5	8	9
↓			↓		
(1)			(2)		

$\xi^{(2)} \xi^{(1)} 6 7 8 9$

$\xi^{(1)} \xi^{(2)} \dots$

$$\begin{aligned} x &= y \\ |x - y| &\leq \epsilon \end{aligned}$$



Aspects of consideration

- ① Business Impact
- ② Tolerance for msg loss
- ③ Storage requirements

Credit Card data Stream → Message delivery semantics

① Business Impact: What is impact of Data loss on Business
Highly critical

② ~~How~~ Tolerance for Data loss :- No we can't lose even 10 min worth of Data

③ Do we have historical data storage:
Requirements?

periodic intervals
—

→ Exactly once